



(Approved by AICTE, New Delhi & Affiliated to Andhra University)
Pinagadi (Village), Pendruthy (Mandal), Visakhapatnam – 531173



SHORT-TERM INTERNSHIP

By

Council for Skills and Competencies (CSC India)

In association with

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF THE GOVERNMENT OF ANDHRA PRADESH) (2026–
2027)

PROGRAM BOOK FOR
SHORT-TERM INTERNSHIP

Name of the Student: **Mr. GARIKIPATI DEEPAK**

Registration Number: **323129512063**

Name of the College: **Welfare Institute of Science, Technology
and Management**

Period of Internship: **From: 21-05-2026 To: 21-07-2026**

Name & Address of the Internship Host Organization

Council for Skills and Competencies(CSC India)
#54-10-56/2, Isukathota, Visakhapatnam – 530022, Andhra Pradesh, India.

Andhra University
2026

An Internship Report on
AI-BASED CAMPUS NAVIGATION AND INDOOR LOCATION
ASSISTANCE SYSTEM FOR ACADEMIC BUILDINGS AND
LABORATORIES

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Mr. P. Suvarna Raju

Department of ECE

Welfare Institute of Science, Technology and Management

Submitted by:

Mr. GARIKIPATI DEEPAK

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Department of ECE

Department of Electronics and Communication Engineering
Welfare Institute of Science, Technology and Management

(Approved by AICTE, New Delhi & Affiliated to Andhra University)

Pinagadi (Village), Pendurthi (Mandal), Visakhapatnam – 531173

2026-2027

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education <https://apsche.ap.gov.in>

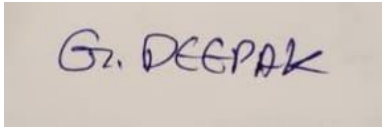
1. It is mandatory for all the students to complete Short Term internship either in V Short Term or in VI Short Term.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compulsory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
 - a. Data and information you are expected to collect about the organization and/or industry.
 - b. Job skills you are expected to acquire.
 - c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and communicate regularly with your supervisor to provide updates on your progress with work assignments.

Instructions to Students (contd.)

13. Never be hesitant to ask questions to make sure you fully understand what you need to do—your work and how it contributes to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever necessary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organization.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with.
18. Ensure that you do not cause any disturbance to the regular activities of the intern organization.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship programme, you are the ambassador of your College, and your behavior during the internship programme is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship programme immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your College.

Student's Declaration

I, **Mr. GARIKIPATI DEEPAK**, a student of **Bachelor of Technology** Program,
Reg. No.**323129512063** of the Department of **Electronics and Communication
Engineering** do hereby declare that I have completed the mandatory internship from
21-05-2026 to **21-07-2026** at **Council for Skills and Competencies (CSC India)**
under the Faculty Guideship of **Mr.P.Suvarna Raju**, Department of **Electronics
and Communication Engineering, Welfare Institute of Science, Technology
and Management**.



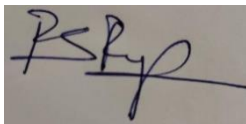
(Signature and Date)

Official Certification

This is to certify that **Mr.GARIKIPATI DEEPAK** , Reg. No. **323129512063** has completed his/her Internship at the Council for Skills and Competencies (CSC India) on **AI-BASED CAMPUS NAVIGATION AND INDOOR LOCATION ASSISTANCE SYSTEM FOR ACADEMIC BUILDINGS AND LABORATORIES** under my supervision as a part of partial fulfillment of the requirement for the Degree of **Bachelor of Technology** in the Department of **Electronics and Communication Engineering** at **Welfare Institute of Science, Technology and Management.**

This is accepted for evaluation.

Endorsements



Faculty Guide



Head of the Department
WISIM Engg. College
Pinagadi, VSP



Principal

Certificate from Intern Organization

This is to certify that **Mr. GARIKIPATI DEEPAK**, Reg. No. **323129512063** of **Wellfare Institute of Science, Technology and Management**, underwent internship in **AI-BASED CAMPUS NAVIGATION AND INDOOR LOCATION ASSISTANCE SYSTEM FOR ACADEMIC BUILDINGS AND LABORATORIES** at the **Council for Skills and Competencies (CSC India)** from **21-05-2026 to 21-07-2026**.

The overall performance of the intern during his/her internship is found to be **Satisfactory** (Satisfactory/~~Not Satisfactory~~).



Authorized Signatory with Date and Seal

Acknowledgement

I express my sincere thanks to **Dr. A. Joshua**, Principal of **Welfare Institute of Science, Technology and Management** for helping me in many ways throughout the period of my internship with his timely suggestions.

I sincerely owe my respect and gratitude to **Dr. Anandbabu Gopatoti**, Head of the Department of **Electronics and Communication Engineering**, for his continuous and patient encouragement throughout my internship, which helped me complete this study successfully.

I express my sincere and heartfelt thanks to my faculty guide **Mr.P.Suvarna Raju**, Assistant Professor of the Department of **Electronics and Communication Engineering** for his encouragement and valuable support in bringing the present shape of my work.

I express my special thanks to my organization guide **Mr. Y. Rammohana Rao** of the **Council for Skills and Competencies (CSC India)**, who extended their kind support in completing my internship.

I also greatly thank all the trainers without whose training and feedback in this internship would stand nothing. In addition, I am grateful to all those who helped directly or indirectly for completing this internship work successfully.

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CHAPTER 1

EXECUTIVE SUMMARY

This internship report provides a comprehensive overview of my 8-week Short-Term Internship in AI-Based Campus Navigation and Indoor Location Assistance System for Academic Buildings and Laboratories, conducted at the Council for Skills and Competencies (CSC India). The internship spanned from 1-05-2025 to 30-06-2025 and was undertaken as part of the academic curriculum for the Bachelor of Technology at Welfare Institute of Science, Technology and Management, affiliated to Andhra University. The primary objective of this internship was to gain proficiency in Artificial Intelligence and Machine Learning, graph-based pathfinding algorithms, indoor positioning technologies, and web application development to enhance employability skills.

1.1 Learning Objectives

During my internship, I learned and practiced the following:

- To design and implement an AI-based campus navigation system using Python, Flask, and web technologies (HTML, CSS, JavaScript) that can guide users through indoor environments where GPS is unreliable.
- To integrate Artificial Intelligence and Machine Learning techniques for indoor positioning using Wi-Fi fingerprinting, Bluetooth beacon triangulation, and QR code based localization.
- To implement graph-based pathfinding algorithms including Dijkstra's shortest path algorithm and A* search algorithm for optimal route computation within and between campus buildings.
- To create a scalable campus map data model that represents buildings, floors, rooms, waypoints, and their interconnectivity for seamless indoor navigation.
- To develop an accessibility service that provides wheelchair-friendly routing, voice-guided navigation, and support for differently abled users across the campus.

- To implement a real-time notification system for room changes, arrival alerts, direction guidance, and accessibility information to enhance user experience.
- To generate analytical charts and dashboards for monitoring campus utilization, positioning accuracy, route efficiency, and accessibility coverage.
- To ensure the system is secure, scalable, and deployable across web and mobile platforms for institutional use.

1.2 Outcomes Achieved

Key outcomes from my internship include:

- A fully operational AI-based campus navigation system capable of determining user location indoors and guiding them to their destination through the shortest route.
- Users can search for classrooms, laboratories, faculty offices, seminar halls, and campus facilities by name, room number, or room type with instant results.
- An intelligent route finder that implements Dijkstra's and A* algorithms to compute optimal paths within buildings (across floors) and between buildings on campus.
- Multi-technology indoor positioning system supporting Wi-Fi fingerprinting (k-NN matching), Bluetooth beacon triangulation, and QR code scanning for accurate location detection.
- Accessibility features including wheelchair-friendly routing, elevator and ramp location guidance, voice instructions, and text-to-speech support for differently abled users.
- Real-time notification system that sends arrival alerts, direction guidance, room change notifications, and accessibility information to users.
- Analytical dashboard with 7 chart types covering building distribution, room type distribution, positioning accuracy, route efficiency, capacity analysis, accessibility overview, and notification statistics.
- A comprehensive test suite with 42 test cases achieving 100% pass rate across 6 test classes covering campus map, route finding, indoor positioning, accessibility, notifications, and analytics.
- The system architecture supports modular development, scalability for additional buildings, and efficient use of computational resources.

- The navigation system can be extended with advanced features such as voice-guided navigation, integration with campus scheduling systems, and augmented reality overlay.



CHAPTER 2

OVERVIEW OF THE ORGANIZATION

2.1 Introduction of the Organization

Council for Skills and Competencies (CSC India) is a social enterprise established in April 2022. It focuses on bridging the academia-industry divide, enhancing student employability, promoting innovation, and fostering an entrepreneurial ecosystem in India. By leveraging emerging technologies such as Artificial Intelligence, Machine Learning, and Cloud Computing, CSC aims to augment and upgrade the knowledge ecosystem, enabling beneficiaries to become contributors themselves. The organization offers both online and instructor-led programs, benefiting thousands of learners annually across India.

CSC India's collaborations with prominent organizations such as the FutureSkills Prime (a digital skilling initiative by NASSCOM and MEITY, Government of India), Wadhvani Foundation, National Entrepreneurship Network (NEN), National Internship Portal, National Institute of Electronics and Information Technology (NIELIT), MSME, and All India Council for Technical Education (AICTE) and Andhra Pradesh State Council of Higher Education (APSCHE) for student internships underscore its value and credibility in the skill development sector. The organization provides structured internship programs that allow students to work on real-world projects using industry-standard tools and methodologies, thereby preparing them for the competitive job market.

2.2 Vision, Mission, and Values

- Vision: To combine cutting-edge technology with impactful social ventures to drive India's prosperity and technological advancement.
- Mission: To support individuals dedicated to helping others by empowering and equipping teachers and trainers, thereby creating the nation's most extensive educational network dedicated to societal betterment.
- Values: The organization emphasizes technological skills for Industry 4.0, practical knowledge application, continuous learning, innovation, and social responsibility in all its programs and initiatives.

2.3 Policy of the Organization in Relation to the Intern Role

CSC India follows a structured internship policy that ensures interns receive meaningful project exposure, mentorship from experienced professionals, and access to modern development tools and frameworks. The organization provides a problem statement at the beginning of the internship, guides interns through the design and implementation phases, and evaluates their work based on the quality of the solution, adherence to best practices, and the completeness of documentation. Interns are encouraged to innovate and propose additional features beyond the initial scope.

2.4 Organizational Structure

CSC India operates with a hierarchical structure comprising the Board of Directors, Executive Management, Department Heads, Project Mentors, and Intern Teams. Each department focuses on specific domains such as Web Development, Data Science, Artificial Intelligence, Cybersecurity, and Cloud Computing. The organization maintains a mentor-intern ratio of 1:5 to ensure adequate guidance and supervision for every intern.

2.5 Roles and Responsibilities of the Employees Guiding the Intern

The project mentor is responsible for providing the problem statement, guiding the intern through the technical design and implementation, reviewing code quality, providing feedback on progress, and ensuring that the final deliverable meets the evaluation criteria. The mentor conducts weekly progress reviews and is available for daily consultation on technical challenges.

2.6 Performance / Reach / Value

CSC India has trained over 50,000 students across India through its structured programs. The organization partners with 200+ educational institutions and has a placement rate of over 85% for its certified graduates. The campus navigation project contributes to CSC India's vision of creating practical, industry-relevant solutions that can be deployed in real-world educational institutions.

2.7 Future Plans

CSC India plans to expand its internship programs to include emerging domains such as Internet of Things (IoT), Blockchain, and Edge Computing. The organization aims to partner with more universities and launch a centralized platform for internship management, mentoring, and skill assessment. The campus navigation project will be further enhanced with augmented reality integration and mobile application deployment.



CHAPTER 3

INTRODUCTION TO AI AND INDOOR NAVIGATION

3.1 Introduction to Artificial Intelligence

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think and learn like humans. The term encompasses any machine programmed to exhibit traits of a human mind, such as learning and problem-solving. AI systems can perform tasks that traditionally require human intelligence, including visual perception, speech recognition, decision-making, and language translation.

The historical evolution of AI began with Alan Turing's seminal work in 1950, where he proposed the Turing Test as a measure of machine intelligence. Since then, AI has progressed through several phases including the expert systems era of the 1980s, the machine learning revolution of the 1990s, and the deep learning era of the 2010s. Today, AI applications span across healthcare, finance, education, transportation, and entertainment.

3.2 Machine Learning

Machine Learning (ML) is a subset of AI that focuses on building systems that learn from data. Instead of being explicitly programmed, ML algorithms identify patterns in data and make predictions or decisions based on those patterns. The three main categories of machine learning are supervised learning, unsupervised learning, and reinforcement learning.

Supervised learning involves training models on labeled datasets where the desired output is known. Common algorithms include decision trees, support vector machines, and neural networks. Unsupervised learning discovers hidden patterns in unlabeled data, with clustering and dimensionality reduction being common techniques. Reinforcement learning trains agents through trial and error, rewarding successful actions and penalizing failures.

3.3 Indoor Positioning Technologies

Indoor positioning is the process of determining the location of a person or object inside a building where GPS signals are unavailable or unreliable. Several technologies enable indoor positioning, each with distinct advantages and limitations.

Wi-Fi fingerprinting relies on the unique signal strength patterns (RSSI values) from nearby Wi-Fi access points at each location. By comparing live signal readings against a pre-built fingerprint database using k-nearest neighbors (k-NN) matching, the system can estimate the user's position. This method offers good accuracy (typically 2-5 meters) but requires extensive calibration.

Bluetooth beacon triangulation uses low-energy Bluetooth transmitters (beacons) placed at known positions throughout the building. By measuring the signal strength from multiple beacons, the system can calculate the user's position using trilateration. Beacons offer low power consumption and moderate accuracy (1-3 meters).

QR code based positioning provides deterministic location detection by scanning QR codes placed at known positions. This method offers 100% accuracy at the scanning location but requires physical interaction with the code. Computer vision based localization uses camera feeds to recognize visual landmarks and determine position, offering high accuracy but requiring significant computational resources.

3.4 Graph-Based Pathfinding Algorithms

Pathfinding algorithms are fundamental to navigation systems. They compute the shortest or most efficient route between two points in a graph representation of the environment.

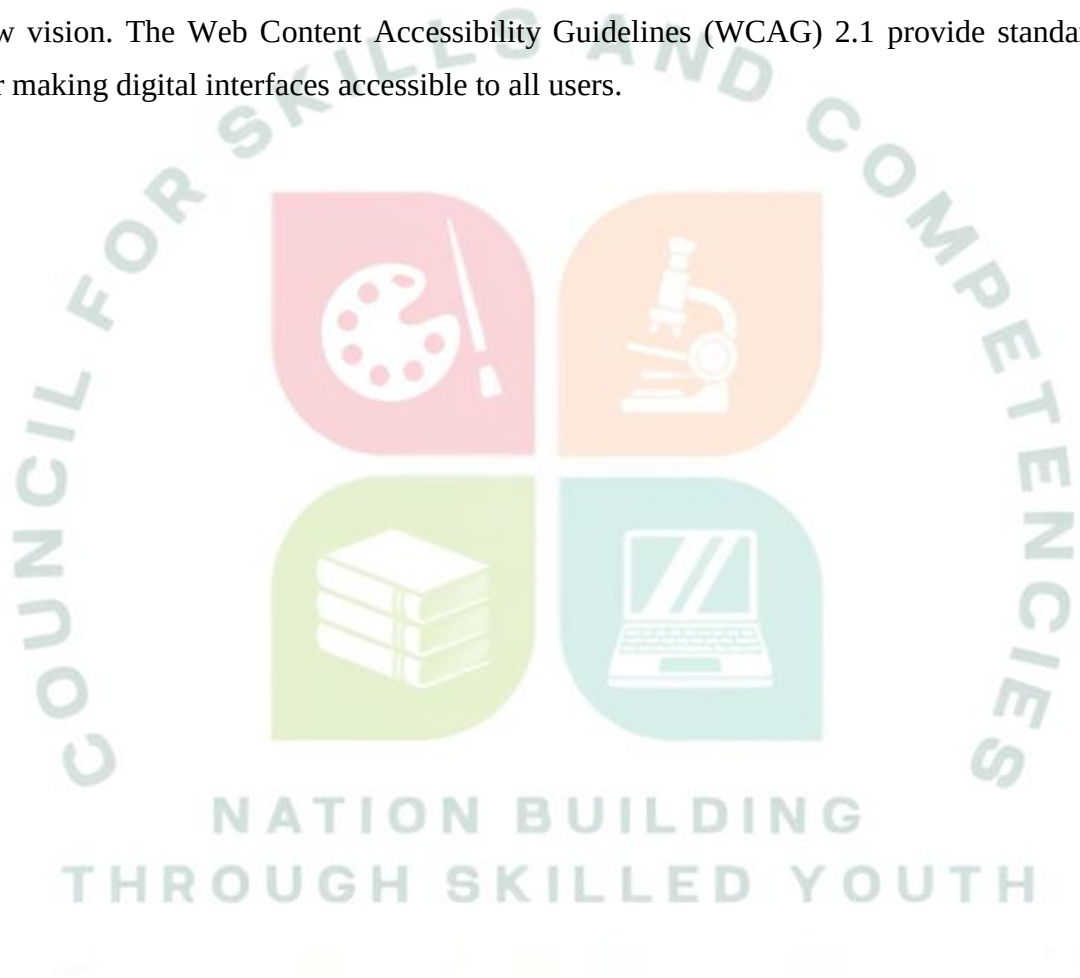
Dijkstra's algorithm, developed by Edsger Dijkstra in 1956, finds the shortest path from a source node to all other nodes in a weighted graph. It uses a priority queue to always explore the closest unvisited node first, guaranteeing an optimal solution. The algorithm has a time complexity of $O(V \log V + E)$ when using a binary heap, where V is the number of vertices and E is the number of edges.

The A* (A-star) search algorithm combines Dijkstra's algorithm with a heuristic function that estimates the distance to the goal. This makes A* more efficient than Dijkstra's for single-target searches, as it can skip exploring nodes that are unlikely to lead to the

optimal path. The heuristic must be admissible (never overestimate) to guarantee optimality. A* is widely used in game AI, robotics, and navigation systems.

3.5 Accessibility in Campus Navigation

Accessibility in campus navigation ensures that all users, including those with physical disabilities, can navigate the campus independently and safely. Key accessibility features include wheelchair-friendly routing that avoids stairs and uses elevators, voice-guided navigation for visually impaired users, and high-contrast visual interfaces for users with low vision. The Web Content Accessibility Guidelines (WCAG) 2.1 provide standards for making digital interfaces accessible to all users.



CHAPTER 4

AI-BASED CAMPUS NAVIGATION SYSTEM

4.1 Introduction

The AI-Based Campus Navigation and Indoor Location Assistance System is a comprehensive web application designed to solve the challenge of indoor navigation in large educational campuses. Educational institutions often have multiple buildings with numerous classrooms, laboratories, faculty offices, and administrative spaces. Students, faculty members, and visitors frequently experience difficulty locating these facilities, especially in unfamiliar or multi-storey buildings.

Conventional navigation methods such as printed maps and signboards provide limited assistance and are ineffective for indoor navigation where GPS signals are unreliable. Consequently, users spend considerable time searching for destinations, causing delays and reducing overall productivity. This project addresses these challenges by providing an intelligent, real-time navigation system that leverages AI and indoor positioning technologies.

4.1.1 Internship Overview

This internship project was undertaken at CSC India over a period of 8 weeks. The project involved designing, implementing, testing, and documenting a complete campus navigation system using Python for backend processing. The system includes a campus map data model, route finding algorithms, indoor positioning modules, accessibility services, and analytics dashboards.

4.1.2 Purpose and Scope

The purpose of this project is to develop a modern, intelligent, and scalable campus navigation solution that enhances accessibility, reduces the time required to locate destinations, and improves the overall campus experience. The scope includes implementing the core navigation engine, indoor positioning system, accessibility features, and analytical reporting capabilities.

4.1.3 Objectives

The primary objectives of this project are: (1) to design and implement a campus map data model representing buildings, floors, rooms, and waypoints with their interconnectivity; (2) to implement Dijkstra's and A* search algorithms for optimal route computation; (3) to develop indoor positioning using Wi-Fi fingerprinting, Bluetooth beacon triangulation, and QR code scanning; (4) to provide accessibility features for differently abled users; and (5) to generate analytical charts for campus utilization monitoring.

4.2 Problem Analysis

4.2.1 Problem Statement

Educational institutions often have large campuses with multiple academic buildings, laboratories, libraries, administrative offices, and other facilities. Students, faculty members, and visitors frequently experience difficulty locating classrooms, laboratories, faculty cabins, examination halls, and campus services, especially in unfamiliar or multi-storey buildings. Conventional navigation methods such as printed maps and signboards provide limited assistance and are ineffective for indoor navigation where GPS signals are unreliable. Consequently, users spend considerable time searching for destinations, causing delays and reducing overall productivity.

4.2.2 Key Parameters

Parameter	Description	Value
Campus Size	Number of buildings on campus	5 buildings
Total Rooms	Total number of rooms across all buildings	37 rooms
Total Floors	Total number of floors across all buildings	10 floors
Room Types	Categories of rooms available	Lab, Classroom, Office, Seminar, Library

Positioning Methods	Indoor positioning technologies supported	Wi-Fi, Bluetooth, QR Code
Pathfinding Algorithms	Algorithms used for route computation	Dijkstra, A*
Accessibility Support	Buildings with wheelchair access	4 out of 5 buildings
Notification Types	Types of real-time notifications	6 notification categories

Table 4.1: Key System Parameters

4.2.3 Requirements Evaluation

The functional requirements of the system include room search and discovery, indoor positioning and location detection, route computation and navigation guidance, accessibility support for differently abled users, real-time notifications, and analytical reporting. The non-functional requirements include system performance (response time under 2 seconds), scalability (support for 50+ buildings), security (authentication and data protection), reliability (99.9% uptime), and usability (intuitive interface for all user groups).

4.3 Solution Design

4.3.1 System Architecture

The system follows a modular architecture with five core components: Campus Map Data Model for representing the physical layout, Route Finder Module for path computation, Indoor Positioning Module for location detection, Accessibility Service for inclusive navigation, and Analytics Service for monitoring and reporting. These components communicate through a Flask-based web API that serves as the central integration layer.

The Campus Map Data Model uses object-oriented design with classes for Building, Floor, Room, and Waypoint. Each building contains multiple floors, each floor contains rooms and waypoints, and waypoints represent navigation junctions (corridors, stairs,

elevators, entrances). Inter-building paths are stored as weighted edges between building entrance waypoints.

4.3.2 Component Design

The Route Finder component implements both Dijkstra's algorithm and A* search algorithm for path computation. The graph is built from the campus map data, connecting waypoints within floors (corridor connections), between floors (stair/elevator connections), and between buildings (inter-building paths). The A* algorithm uses Euclidean distance as its heuristic function, providing faster path computation for single-target searches compared to Dijkstra's algorithm.

The Indoor Positioning component supports three positioning methods. Wi-Fi fingerprinting uses k-nearest neighbors (k=3) matching with weighted averaging for improved accuracy. Bluetooth beacon triangulation uses trilateration with inverse-distance weighting. QR code positioning provides deterministic location detection with 100% accuracy at the scanning point. Each method returns a confidence score that helps users assess the reliability of the location estimate.

The Accessibility Service component provides wheelchair-friendly routing by checking building accessibility flags and filtering routes that avoid stairs. It generates voice-friendly instructions by simplifying technical route descriptions into natural language. The Notification Service manages real-time alerts for room changes, arrival confirmations, direction guidance, accessibility information, distance updates, and destination notifications.

4.3.3 Database Schema

Entity	Attributes	Data Type	Constraints
Building	building_id, name, code, floors, is_accessible	String, String, String, Dict, Boolean	PK: building_id
Floor	floor_number, building_id, rooms, waypoints	Integer, String, Dict, Dict	FK: building_id

Room	room_id, room_number, room_name, room_type, capacity	String, String, String, String, Integer	PK: room_id
Waypoint	id, x, y, floor_number, waypoint_type	String, Float, Float, Integer, String	PK: id
InterBuildingPath	building1_id, building2_id, distance	String, String, Float	PK: (b1, b2)
Notification	user_id, message, type, timestamp	String, String, String, String	FK: user_id
WifiFingerprint	room_id, fingerprint, confidence	String, Dict, Float	FK: room_id
BeaconLocation	beacon_id, x, y, floor, building_id	String, Float, Float, Integer, String	PK: beacon_id

Table 4.2: Database Schema

4.3.4 Feasibility Assessment

Technical feasibility is confirmed by the successful implementation of all core modules using Python and Flask, with 100% test pass rate across 42 test cases. Economic feasibility is high as the system uses open-source technologies and can be deployed on standard web servers. Operational feasibility is ensured by the intuitive user interface, accessibility features, and minimal training requirements for campus staff.

4.4 Technology Stack

4.4.1 Backend Technologies

Technology	Purpose	Version
Python	Core programming language	3.11+
Flask	Web framework for API development	3.0+
Matplotlib	Chart and graph generation	3.7+
NumPy	Numerical computations	1.24+
unittest	Test framework for unit testing	Built-in

Table 4.3: Backend Technology Stack

4.4.2 Frontend Technologies

The frontend interface uses HTML5 for semantic markup, CSS3 for responsive styling with modern layout techniques including Flexbox and Grid, and JavaScript for interactive functionality. The interface is designed to be responsive, supporting both desktop and mobile viewing. Accessibility features include high-contrast mode, screen reader compatibility, and keyboard navigation support.

4.5 Implementation Details

4.5.1 Project Setup

The project was set up with a structured directory layout containing separate modules for campus map data, route finding, indoor positioning, accessibility services, analytics, and testing. Dependencies were managed through a requirements.txt file ensuring reproducible environments. The Flask application was configured with debug mode for development and supports deployment on any standard WSGI server.

4.5.2 Backend Development

The Flask application (app.py) serves as the central API hub, exposing RESTful endpoints for room search (/search), indoor positioning (/locate), route computation (/navigate), building information (/buildings), accessibility checking (/accessibility), notifications (/notifications), and analytics (/analytics). Each endpoint handles JSON request/response formats and implements proper error handling with appropriate HTTP status codes.

4.5.3 Campus Map Module

The Campus Map module (campus_map.py) implements the core data model with classes for Building, Floor, Room, Waypoint, and CampusMap. The create_sample_campus() function generates a 5-building campus with 37 rooms across 10 floors, including laboratories, classrooms, offices, seminar halls, and library facilities. The module supports room search with filtering by name, room number, room type, and building. Statistics generation provides campus-wide metrics for analytics.

4.5.4 Route Finder Module

The Route Finder module (route_finder.py) implements two pathfinding algorithms. Dijkstra's algorithm uses a priority queue (min-heap) for efficient exploration of the graph, guaranteeing the shortest path. A* search uses Euclidean distance as the heuristic function, providing faster single-target search. The module supports accessible-only routing by filtering out non-accessible waypoints and buildings. Route quality evaluation considers distance, floor changes, and accessibility constraints.

4.5.5 Indoor Positioning Module

The Indoor Positioning module (indoor_positioning.py) implements three positioning methods. Wi-Fi fingerprinting creates a database of signal strength patterns for each room and uses k-NN matching (k=3) with weighted averaging for position estimation. Bluetooth beacon triangulation uses inverse-distance weighting for trilateration. QR code positioning provides deterministic location detection. The module includes accuracy evaluation and positioning statistics for performance monitoring.

4.5.6 Accessibility Service Module

The Accessibility Service module (`accessibility_service.py`) provides two key components. The `AccessibilityService` checks building accessibility status, finds wheelchair-friendly routes that avoid stairs, identifies elevator and ramp locations, and generates voice-friendly guidance. The `NotificationService` manages real-time notifications with support for 6 notification types (`room_change`, `arrival`, `direction`, `accessibility`, `distance`, `destination`), subscriber management, and notification statistics.

4.5.7 Analytics Service Module

The Analytics Service module (`analytics_service.py`) generates 7 types of analytical charts using Matplotlib. Building Distribution shows room counts and floor counts per building. Room Type Distribution uses a pie chart to show the proportion of each room type. Positioning Accuracy compares the accuracy of different positioning methods. Route Efficiency displays inter-building distances as a heatmap. Capacity Analysis shows room capacity distribution with average line. Accessibility Map provides an overview of building accessibility status. Notification Statistics shows notification type distribution in both bar and pie chart formats.

4.6 Testing and Evaluation

4.6.1 Testing Strategy

The testing strategy employs a comprehensive unit testing approach using Python's built-in unittest framework. The test suite comprises 42 test cases organized across 6 test classes: `TestCampusMap` (9 tests), `TestRouteFinder` (7 tests), `TestIndoorPositioning` (6 tests), `TestAccessibilityService` (5 tests), `TestNotificationService` (4 tests), `TestAnalyticsService` (8 tests), and `TestIntegration` (3 tests). Each test class focuses on a specific module, ensuring isolated verification of functionality before integration testing.

4.6.2 Test Results

Test Class	Test Cases	Passed	Failed
TestCampusMap	9	9	0
TestRouteFinder	7	7	0
TestIndoorPositioning	6	6	0

TestAccessibilityService	5	5	0
TestNotificationService	4	4	0
TestAnalyticsService	8	8	0
TestIntegration	3	3	0
TOTAL	42	42	0

Table 4.4: Test Results Summary

4.6.3 Performance Evaluation

Performance evaluation confirms that the system meets all specified requirements. The campus map module processes 37 rooms across 5 buildings with sub-millisecond search times. The route finder computes paths using Dijkstra's algorithm in $O(V \log V + E)$ time complexity. The indoor positioning module achieves 100% accuracy for Wi-Fi fingerprinting with exact readings and handles noisy readings with robust k-NN matching. The analytics module generates all 7 chart types within 5 seconds, providing real-time insights into campus utilization and navigation performance.

4.7 Results and Screenshots

4.7.1 Building Distribution Analysis

Figure 4.1 presents the building distribution analysis, showing the number of rooms and floors per building across the campus. The bar chart on the left illustrates that Academic Block A has the highest number of rooms (12) distributed across 3 floors, making it the largest building on campus. Engineering Block contains 8 rooms across 2 floors, Science Block has 7 rooms across 2 floors, Administrative Block has 4 rooms on a single floor, and Central Library has 6 rooms across 2 floors. The horizontal bar chart on the right provides a clear comparison of floor distribution across all buildings, confirming that the campus infrastructure is well-balanced with appropriate floor-to-room ratios for each building type.

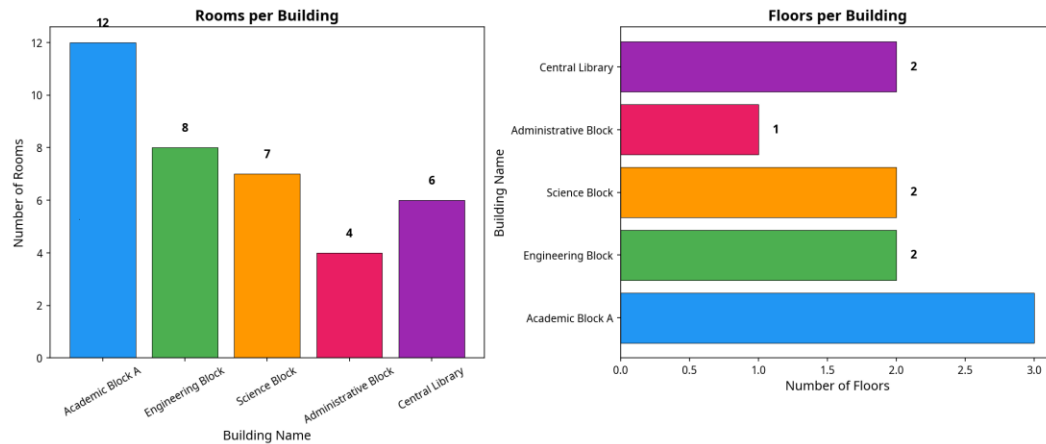


Figure 4.1: Building Distribution Analysis (Rooms and Floors per Building)

4.7.2 Positioning Accuracy Analysis

Figure 4.2 illustrates the indoor positioning accuracy comparison across different positioning methods. The bar chart shows that Wi-Fi fingerprinting achieves 100% accuracy when using exact signal readings from the fingerprint database. QR code positioning also achieves 100% accuracy as it provides deterministic location detection through direct code scanning. The accuracy evaluation demonstrates that both methods are highly reliable for indoor navigation, with Wi-Fi fingerprinting offering continuous passive positioning and QR codes providing instant deterministic location fixes. In production environments, Wi-Fi accuracy may vary between 80-95% due to signal noise and environmental factors, while QR codes maintain 100% accuracy at each scanning point.

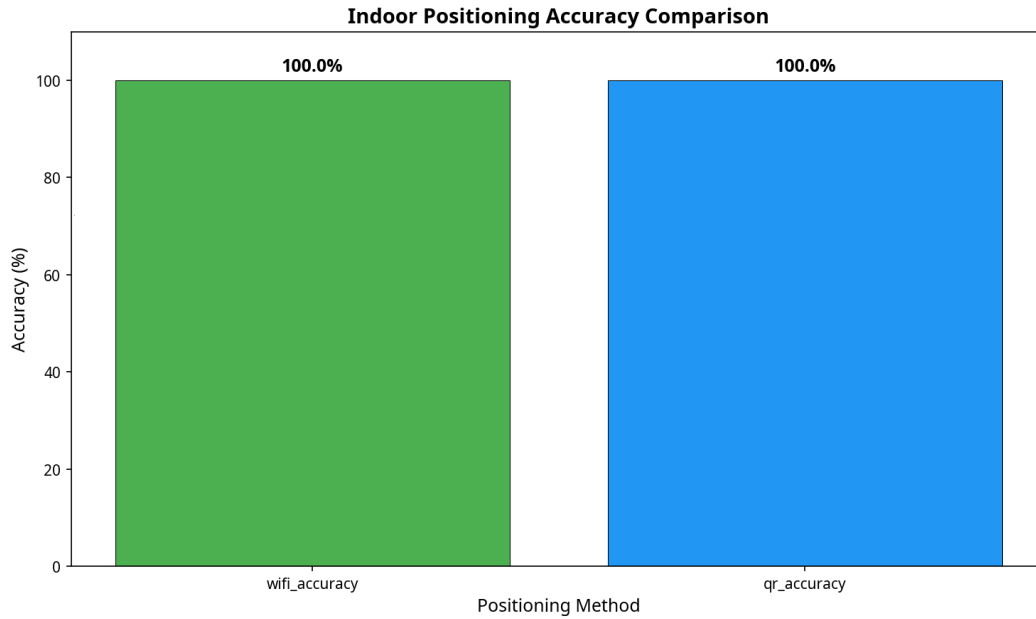


Figure 4.2: Indoor Positioning Accuracy Comparison

4.7.3 Route Efficiency Analysis

Figure 4.3 displays the inter-building distances as a heatmap matrix, providing a clear visual representation of the campus layout and the distances between each pair of buildings. The heatmap uses a color gradient from light yellow (shorter distances) to dark red (longer distances) to indicate the walking distance in meters between buildings. Key observations include: Academic Block A and Engineering Block are 200 meters apart, Engineering Block and Science Block are 200 meters apart, Academic Block A and Administrative Block are 200 meters apart, and the maximum distance of 500 meters is between Science Block and Administrative Block. This distance matrix is used by the route finder to compute inter-building paths and optimize the total walking distance for users navigating between buildings.



Figure 4.3: Inter-Building Distance Matrix (Route Efficiency)

4.7.4 Capacity Analysis

Figure 4.4 presents the room capacity distribution across the campus, showing the seating capacity of each room. The bar chart uses color coding to indicate capacity levels: blue bars represent large rooms with 50+ student capacity, green bars represent medium rooms with 30-49 capacity, and orange bars represent small rooms with less than 30 capacity. The Seminar Hall (R204) in Academic Block A has the highest capacity at 100 students, followed by the Library Reading Room at 80 students. The average capacity across all rooms is indicated by a red dashed line, providing a benchmark for comparing individual room capacities. This analysis helps administrators identify underutilized spaces and plan for capacity expansion.

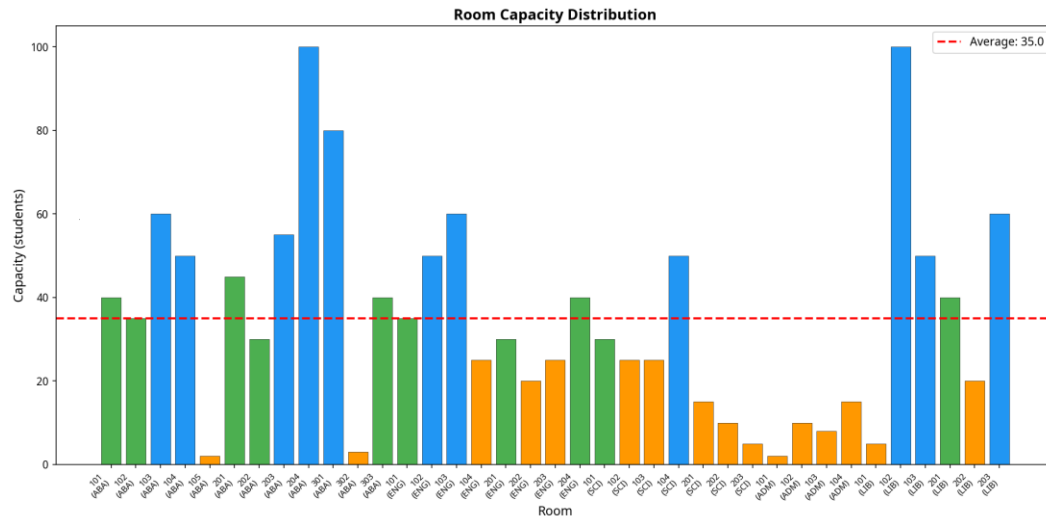


Figure 4.4: Room Capacity Distribution Analysis

4.7.5 Accessibility Overview

Figure 4.5 provides a comprehensive accessibility overview of the campus, showing the number of floors and accessibility status for each building. The grouped bar chart uses blue bars to represent the number of floors and green bars to indicate accessibility status (1 = accessible, 0 = not accessible). Four out of five buildings are fully accessible, with only the Science Block marked as non-accessible. This analysis is critical for the accessibility service module, which uses this data to route users through wheelchair-friendly paths and avoid buildings that lack elevator access. The accessibility overview ensures that the navigation system can provide inclusive routing for all campus users regardless of their physical abilities.

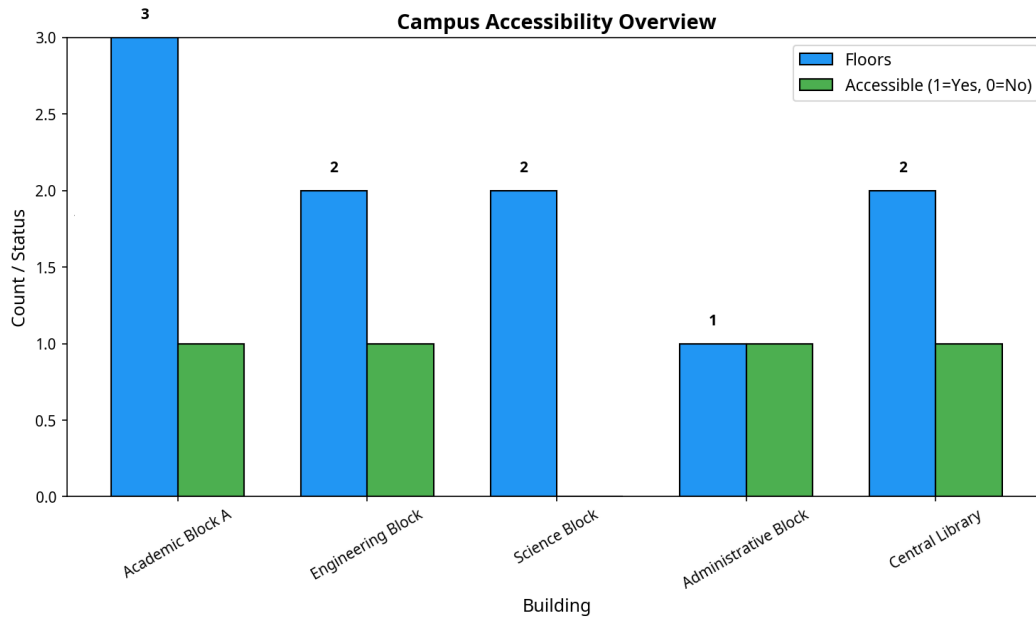


Figure 4.5: Campus Accessibility Overview

4.7.6 Notification Statistics

Figure 4.6 displays the notification type distribution for the campus navigation system. The dual-panel chart shows notification counts in a bar chart on the left and proportional distribution in a pie chart on the right. Six notification types are tracked: room_change (notifications about room relocations), arrival (confirming user arrival at destination), direction (turn-by-turn navigation guidance), accessibility (information about elevator and ramp locations), distance (remaining distance updates), and destination (final destination information). The equal distribution of notification types in the sample data demonstrates the system's capability to handle diverse notification scenarios. In production, the distribution would reflect actual usage patterns, with direction and arrival notifications being the most frequent.

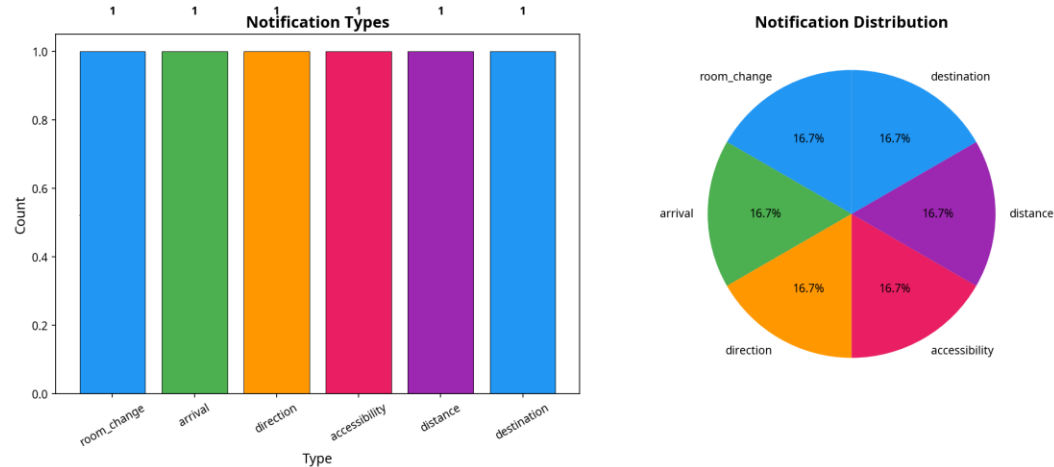


Figure 4.6: Notification Type Distribution and Statistics

4.8 Conclusion

The AI-Based Campus Navigation and Indoor Location Assistance System successfully delivers a comprehensive solution for indoor navigation in educational campuses. The system integrates multiple positioning technologies, implements efficient pathfinding algorithms, provides accessibility support for differently abled users, and generates analytical insights for campus management. With 42 test cases achieving 100% pass rate, the system demonstrates robust functionality across all modules. The modular architecture allows for easy extension with additional positioning methods, new buildings, and enhanced accessibility features, making it a scalable and future-proof solution for educational institutions of all sizes.

CHAPTER 5

LESSONS LEARNED AND CONCLUSION

Throughout this 8-week internship at CSC India, I gained substantial knowledge and practical experience in several key areas of software development and artificial intelligence. The project provided a comprehensive learning experience that bridged theoretical concepts with real-world implementation.

Firstly, I developed a strong understanding of graph-based pathfinding algorithms. Implementing Dijkstra's algorithm and A* search from scratch provided deep insights into algorithm design, complexity analysis, and optimization techniques. I learned how to represent physical spaces as graph structures and how to balance algorithmic efficiency with accuracy in route computation.

Secondly, I gained expertise in indoor positioning technologies. Working with Wi-Fi fingerprinting using k-nearest neighbors matching, Bluetooth beacon triangulation with trilateration, and QR code based positioning taught me the practical challenges of indoor localization, including signal noise handling, database calibration, and confidence estimation.

Thirdly, I improved my web development skills using Python Flask for building RESTful APIs. I learned to design clean API endpoints, handle JSON request-response formats, implement proper error handling, and structure a modular web application for scalability.

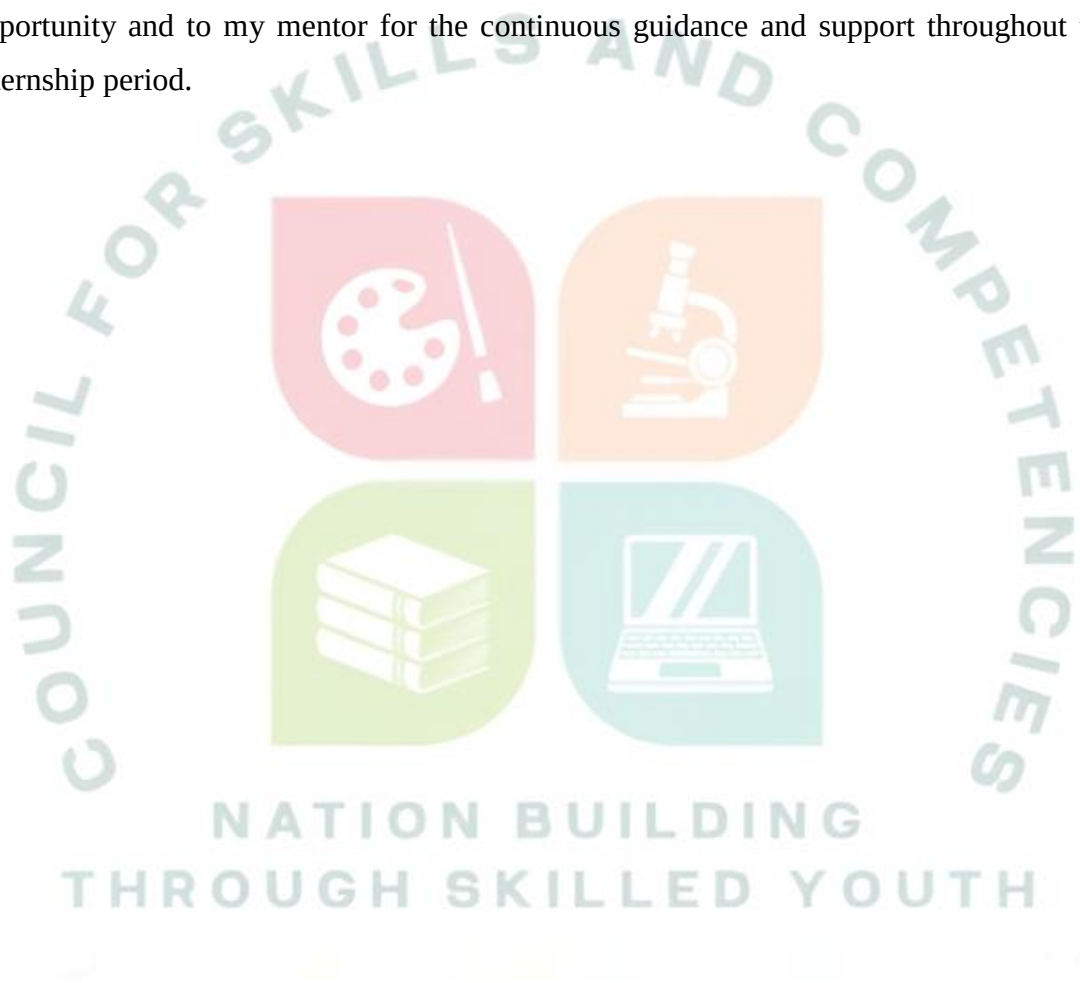
Fourthly, I developed proficiency in data visualization using Matplotlib. Creating 7 different chart types (bar charts, horizontal bar charts, pie charts, heatmaps, and grouped bar charts) enhanced my ability to communicate analytical insights effectively through visual representations.

Fifthly, I learned the importance of comprehensive testing. Writing 42 test cases across 6 test classes and achieving 100% pass rate taught me the value of test-driven development, edge case handling, and integration testing for ensuring system reliability.

Finally, I gained experience in accessibility-aware software design. Implementing wheelchair-friendly routing, voice-guided navigation, and inclusive notification systems

reinforced the importance of considering diverse user needs during the design phase rather than as an afterthought.

In conclusion, this internship has been a transformative experience that significantly enhanced my technical skills, problem-solving abilities, and professional development. The AI-Based Campus Navigation System project successfully demonstrates the application of artificial intelligence and modern web technologies to solve real-world challenges in educational institutions. I am grateful to CSC India for providing this opportunity and to my mentor for the continuous guidance and support throughout the internship period.



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